OFFICIAL INSTITUTIONAL SPECIFICATION

Product Requirements Document (PRD)

VestraAI (Satvam Invest) Quantitative Platform

Version: 2.4.0-PROD **Status:** Approved / Active **Release Date:** October 2026

Markets: US Equities (S&P 500 / NASDAQ) & Indian Equities (NSE / NIFTY 50)

Product Requirements Document (PRD)

VestraAI / Satvam Invest: Autonomous Global Investing Intelligence Platform

Document Information

- **Product Name:** VestraAI (Satvam Invest)
- **Document Version:** 1.0.0
- **Document Status:** Complete / Ready for Production Review
- **Owner:** Product Management & Quantitative Strategy Team
- **Target Release:** Q4 2026 - Q1 2027

1. Executive Summary & Product Vision

1.1 Product Vision

VestraAI is an AI-native financial intelligence platform that democratizes institutional-grade quantitative investing for modern retail investors, family offices, and active wealth managers. By unifying **US Equities (S&P 500 / NASDAQ)** and **Indian Equities (NSE / NIFTY 50)** into a cohesive, multi-currency command center, VestraAI bridges the gap between complex algorithmic hedge-fund models and accessible, actionable portfolio execution.

1.2 Core Value Proposition

- **Institutional Multi-Factor Rigor:** Eliminates emotional trading by evaluating assets through an explainable **5-Module Ensemble Engine** (Factor 35%, XGBoost ML 25%, FinBERT Sentiment 20%, Live VWAP/Momentum 10%, LLM Moat 10%).
- **Live Market Outlier Capture:** Identifies intraday market momentum and breakout volume via real-time **Live Market Movers** screeners.

- **Dual-Bucket Autonomous Portfolio:** Protects long-term wealth through an untouchable **Core Bucket (100% user-directed)** while optimizing opportunistic upside via an autonomous **Tactical Vault (zero-intervention rebalancing)**.
- **Cross-Border Intelligence:** Native dual-currency ($ USD and ₹ INR) engine supporting seamless asset tracking across global markets.

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2. Problem Statement & Market Opportunity

Current Market Pain Points

Information Overload: Noise from social media, biased analyst opinions, and lagging reports.

Black-Box AI Models: Generic LLM responses with zero mathematical explainability or factor rigor.

Emotional Bias: 80%+ retail underperformance caused by panic selling and FOMO chasing.

Fragmented Platforms: Disconnected tools for US and Indian equities with no unified risk view.

VS

VestraAI Platform Solutions

Transparent 5-Module Math: Strict multi-factor scoring (Factor 35%, ML 25%, NLP 20%, VWAP 10%, Moat 10%).

Live Outlier Detection: Real-time screener integration tracking volume breakouts and VWAP crossovers.

Autonomous Dual-Bucket: Core long-term assets untouched; Tactical Vault dynamically rebalanced.

Unified Cross-Border: Native dual-currency terminal seamlessly switching between USD ($) and INR (₹).

2.1 The Retail-Institutional Divide

1. **Information Asymmetry:** Quantitative hedge funds deploy high-frequency sentiment models (FinBERT), alternative data, and low-latency order book analysis. Retail investors rely on lagging quarterly reports and social media hype.
2. **Emotional Execution:** More than 80% of retail traders underperform benchmark indices due to panic-selling during macroeconomic volatility and chasing overbought peaks.
3. **Siloed Ecosystems:** Investors managing portfolios across US and Indian growth stocks must juggle disconnected brokerage platforms, conversions, and news sources without a unified portfolio risk model.

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3. User Personas & Target Demographics

Persona A: The Systematic Retail Trader ("Arjun")

- **Profile:** 29-year-old software engineer and active equity trader based in Bengaluru / San Francisco.
- **Goals:** Wants data-driven buy signals, clear entry target prices, and probability metrics rather than generic market tips.
- **Pain Point:** Lacks time during market hours to run Python backtests and scan hundreds of stock charts across US and NSE markets.
- **Key Feature Used: Top 3 Algo Picks** and **Live Market Movers Scanner**.

Persona B: The Long-Term Wealth Compounder ("Sarah")

- **Profile:** 42-year-old corporate executive building generational wealth across US Tech leaders and Indian infrastructure.

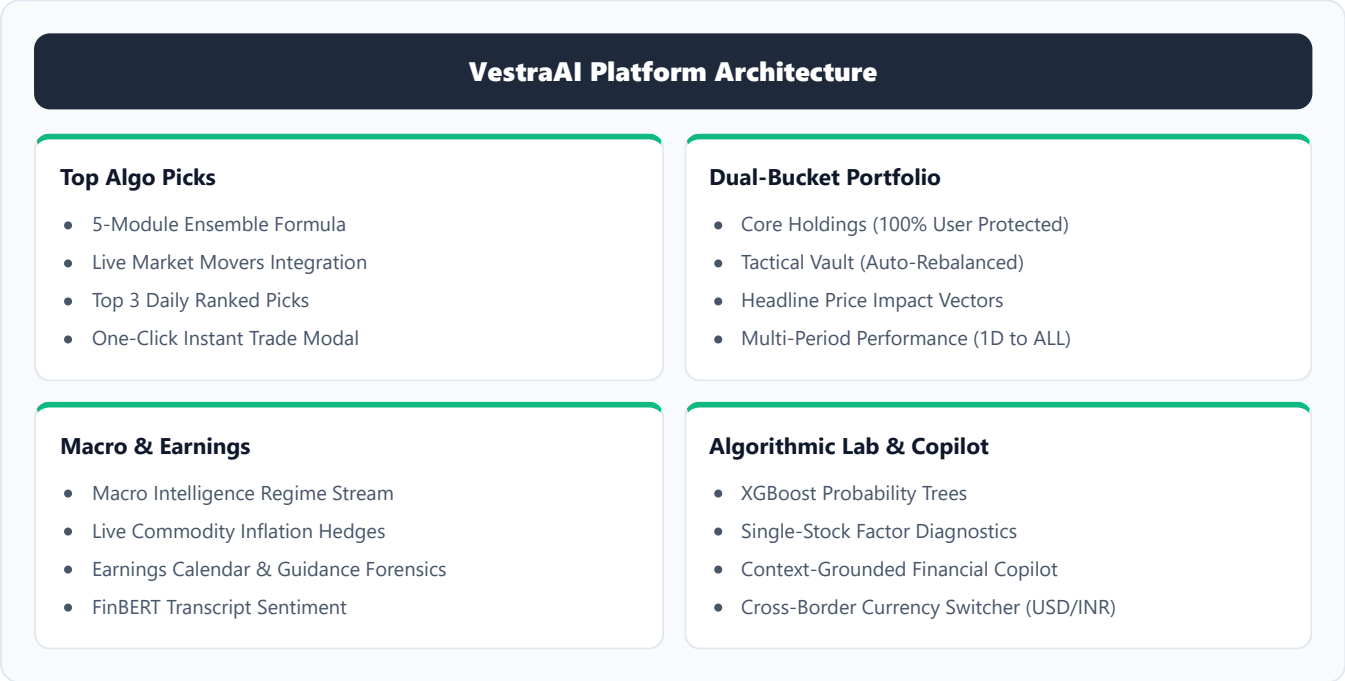
- **Goals:** Wants capital preservation, strong free cash flow companies, and automated risk management without constant monitoring.
- **Pain Point:** Anxiety over geopolitical shocks and inflation destroying portfolio purchasing power.
- **Key Feature Used: Dual-Bucket Portfolio (Core Protected)** and **Macro Intelligence Stream**.

Persona C: The Fundamental & Macro Allocator ("Devon")

- **Profile:** 35-year-old RIA (Registered Investment Advisor) / Family Office Analyst.
- **Goals:** Requires institutional-grade factor analysis, Put/Call open interest ratios, and transcript forensics to justify allocations to clients.
- **Pain Point:** Traditional Bloomberg / FactSet terminals are prohibitively expensive and lack native generative AI synthesis.
- **Key Feature Used: Earnings Agent**, **Algorithmic Trading Lab**, and **AI Copilot**.

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4. Key Product Epics & Functional Requirements



Epic 1: Top Algo Picks & Live Market Movers Engine

Functional Requirements:

- **FR-1.1 Real-Time Dynamic Universe Scan:** The system must evaluate a continuous universe of 55+ US equities and 52 NIFTY 50 Indian equities.
- **FR-1.2 Dynamic Live Mover Integration:** The engine must automatically ingest real-time `day_gainers` and `most_actives` screeners, filtering for liquid US equities ($P \geq$ |10.00$), and dynamically merging them into the candidate pool.
- **FR-1.3 5-Module Score Computation:** The system must compute and display individual module breakdowns:

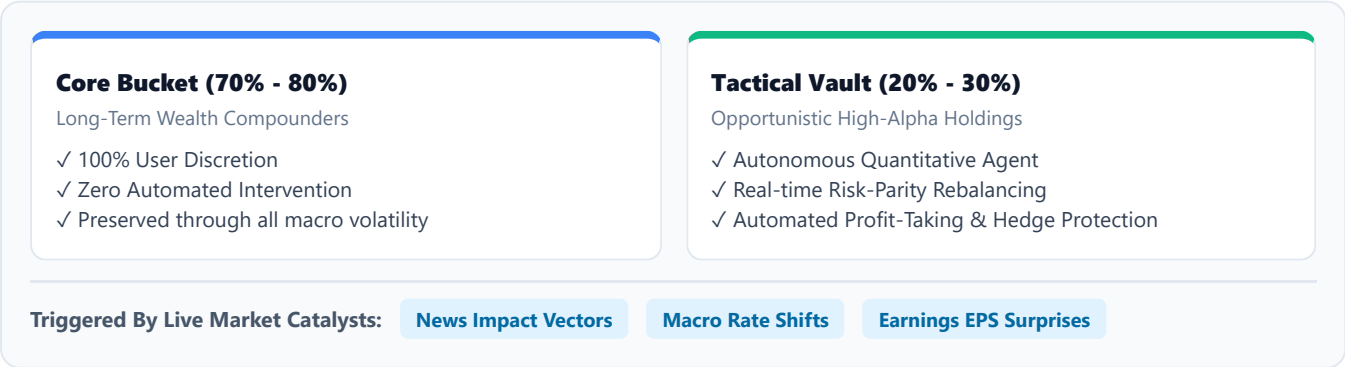
- Factor Score (35%)
- XGBoost Win Probability (25%)
- FinBERT Sentiment Score (20%)
- Live VWAP / Momentum Score (10%)
- LLM Moat Qualitative Score (10%)
- **FR-1.4 4-Layer Conviction Matrix Display:** Each top recommendation card must show sub-scores for:
- Macro & Central Bank Regime (25%)
- Derivatives Structure & PCR (25%)
- Quant & Technical Momentum (30%)
- Fundamental & Earnings (20%)
- **FR-1.5 Dynamic Target & Upside Projection:** Provide a 12-month algorithmic target price (P_{target}) and estimated percentage upside based on the conviction formula.
- **FR-1.6 One-Click Instant Trade Execution:** Users must be able to trigger an instant buy modal directly from any Top Pick card, selecting order quantity and target bucket (Core vs Tactical).

Epic 2: Algorithmic Trading Lab & Single-Stock Deep Dive

Functional Requirements:

- **FR-2.1 Ticker Search & Direct Link Routing:** Users can search any supported ticker or arrive via URL query parameter (e.g., `/algo?ticker=NVDA`) from other pages.
- **FR-2.2 Feature Tree Visualization:** Display XGBoost historical pattern recognition, RSI(14), MACD status, and technical breakout zones.
- **FR-2.3 Qualitative AI Thesis:** Render an automated qualitative investment thesis explaining fundamental tailwinds, competitive moat, and potential risk factors.
- **FR-2.4 Action Badges:** Categorize signals into five distinct tiers: `Strong Buy` (>85), `Buy` ($70\text{-}85$), `Hold` ($55\text{-}70$), `Reduce` ($40\text{-}55$), and `Avoid` (<40).

Epic 3: Autonomous Dual-Bucket Portfolio Architecture



Functional Requirements:

- **FR-3.1 Dual-Bucket Isolation:** Segregate user portfolio into:

- **Core Bucket:** Long-term blue chips, untouchable by automated rebalancing rules.
- **Tactical Vault:** High-alpha, opportunistic holdings managed with automated risk-parity rules.
- **FR-3.2 Portfolio News & Price Impact Vectors:** Ingest financial news and tag holdings with positive, neutral, or negative price impact vector alerts.
- **FR-3.3 Multi-Period Performance Visualization:** Render interactive Recharts performance curves across 1D, 1W, 1M, 1Y, and ALL timeframes.
- **FR-3.4 Local Persistence:** Persist all holdings, transactions, and cash balance to browser storage to maintain state across sessions.

Epic 4: Macro Intelligence Stream & Commodity Hedges

Functional Requirements:

- **FR-4.1 Macro Regime Detection:** Stream real-time synthesized macro themes (inflation sticky vs cooling, Fed rate trajectory, RBI liquidity, geopolitical supply chain shocks).
- **FR-4.2 Real-Time Commodity Benchmarks:** Live price tracking of inflation and energy hedges:
- Gold Futures (`GC=F`) & SPDR Gold (`GLD`)
- Silver Futures (`SI=F`) & iShares Silver (`SLV`)
- WTI Crude Oil (`CL=F`) & Brent Crude (`BZ=F`)
- Copper Futures (`HG=F`), Platinum (`PL=F`), Natural Gas (`NG=F`)
- **FR-4.3 AI Asset Allocation Recommendation:** Interactive modal allowing users to generate an optimal risk-parity allocation (e.g., 40% Chips/Memory, 30% Gold, 20% Crude Oil, 10% Cloud/E-Commerce) based on current macro headwinds.

Epic 5: Autonomous Earnings Intelligence Agent

Functional Requirements:

- **FR-5.1 Upcoming Earnings Calendar:** Filter earnings dates and consensus EPS estimates for both US and Indian holdings.
- **FR-5.2 Transcript Sentiment Forensics:** Score CEO/CFO conference call remarks using FinBERT polarity models.
- **FR-5.3 Guidance Catalyst Scoring:** Flag forward revenue guidance revisions and enterprise deal pipeline momentum.

Epic 6: AI Financial Copilot

Functional Requirements:

- **FR-6.1 Portfolio-Grounded Context:** Automatically inject current cash balance, portfolio holdings, active currency, and top algo picks into the prompt context.
- **FR-6.2 Multi-Provider LLM Fallback:** Support seamless switching between Google Gemini, Anthropic Claude, and OpenAI models.

- **FR-6.3 Financial Query Understanding:** Accurately explain technical terms, calculate risk-reward ratios, and synthesize complex multi-factor metrics into conversational prose.

Epic 7: Global Multi-Market Terminal (US & India)

Functional Requirements:

- **FR-7.1 One-Click Currency Switcher:** Global navbar toggle instantly switches UI display between `$ USD` and `₹ INR`.
- **FR-7.2 Real-Time FX Conversion:** Automatically convert portfolio asset valuations based on live USD/INR exchange rates ($1\ USD \approx 87.5\ INR$).
- **FR-7.3 Unified Search & Quotes:** Single unified search bar capable of resolving US tickers (`NVDA`, `AAPL`) and Indian NSE tickers (`RELIANCE`, `TCS`, `TATAMOTORS`).

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5. User Journey & Workflow Specifications

Journey 1: Morning Briefing & Instant Execution

Architecture Diagram Specification

```
sequenceDiagram
    autonumber
    actor Investor as User
    participant Dash as Dashboard UI
    participant Algo as TopAlgoPicks Component
    participant Modal as Instant Trade Modal
    participant Context as PortfolioContext

    Investor->>Dash: Opens VestraAI Dashboard
    Dash->>Algo: Fetches Top 3 Daily Picks for Active Market (US)
    Algo-->>Dash: Displays #1 Pick (e.g., GOOGL, 88.7% Score)
    Investor->>Algo: Reviews 4-Layer Conviction & XGBoost Win Prob (84.2%)
    Investor->>Algo: Clicks "Buy #1 Pick"
    Algo->>Modal: Opens Trade Execution Dialog
    Investor->>Modal: Selects 5 Shares, assigns to "Core Bucket"
    Investor->>Modal: Clicks "Confirm Execution"
    Modal->>Context: Deducts Cash Balance & Adds 5 Shares of GOOGL
    Context-->>Dash: Portfolio Valuation & Allocation Pie Chart Update
```

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6. User Experience & Design Guidelines

6.1 Visual Aesthetics & Design System

- **Theme:** Modern Cyber-Institutional Dark Mode.
- **Background:** Deep slate (`#0b0f19` to `#0f172a`) with subtle ambient radial gradients.
- **Panels:** Glassmorphism styling (`backdrop-blur-md`, subtle translucent white borders `border-white/10`).

- **Typography:** Inter / SF Pro Display for body text; JetBrains Mono for prices, percentages, and ticker badges.
- **Color Semantics:**
- **Emerald Green (`#10b981`):** Positive returns, Strong Buy, High Conviction, Tailwinds.
- **Electric Blue (`#3b82f6`):** Algorithmic metrics, Derivatives/PCR, US Market tags.
- **Purple / Indigo (`#8b5cf6`):** AI Synthesizer, LLM Moat, Quantum technicals.
- **Amber / Orange (`#f59e0b`):** Warning, Neutral Guidance, High Risk Tactical allocations.
- **Rose Red (`#ef4444`):** Losses, Negative Guidance, Headwinds, De-Risk signals.

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7. Non-Functional Requirements (NFRs)

Category	Requirement	Target Metric
Performance	API Response Time (Cached)	$< 50\ ms$
Performance	Time to Interactive (TTI)	$< 1.2\ s$ on 4G / broadband
Data Freshness	Live Market Movers Cache TTL	$\leq 45\ seconds$
Data Freshness	Background Universe Refresher	Runs every $28\ seconds$
Availability	Server Uptime	99.9% with resilient mock fallbacks
Resilience	Upstream API Throttling Tolerance	Zero user-visible crashes; immediate institutional baseline fallback
Compatibility	Browser Support	Chrome, Edge, Safari, Firefox (last 2 major versions)

8. Success Metrics & Key Performance Indicators (KPIs)

Architecture Diagram Specification

```
graph LR
    subgraph AlgoKPIs ["1. Quantitative Performance"]
        Alpha["Alpha Generation vs S&P 500 / Nifty 50 (> +4.5% annualized)"]
        Sharpe["Sharpe Ratio (> 1.85)"]
        MaxDD["Maximum Drawdown (< -12.0%)"]
    end

    subgraph UserKPIs ["2. Product Engagement"]
        DAU["Daily Active Users (DAU) & Stickiness (DAU/MAU > 40%)"]
        TopPickClicks["Top Algo Pick Conversion (> 22% of views lead to trade modal)"]
        CopilotChats["AI Copilot Inquiries per Active Session (> 3.5 queries)"]
    end

    subgraph ReliabilityKPIs ["3. System Reliability"]
        Latency["P95 API Latency (< 120ms)"]
        CacheHit["Cache Hit Ratio (> 94%)"]
    end
```

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9. Product Roadmap

Phase 1: MVP & Foundation (Current Milestone - Completed)

- ✅ 5-Module Quantitative Ensemble Engine implementation.
- ✅ Dynamic Live Market Movers scanner with screener integration.
- ✅ 4-Layer Institutional Quantitative Matrix.
- ✅ Top 3 Algo Picks for both US (S&P 500) and Indian (NIFTY 50) markets.
- ✅ Dual-Bucket Portfolio architecture (Core vs Tactical Vault).
- ✅ Macro Intelligence Stream and real-time Commodities tracking.
- ✅ AI Financial Copilot and Earnings Agent.

Phase 2: Live Broker Integrations & Automated Execution (Q1 - Q2 2027)

- 🔄 Direct OAuth2 Broker APIs:
- **US:** Interactive Brokers (IBKR), Alpaca, Charles Schwab.
- **India:** Zerodha Kite Connect, Groww, Angel One.
- 🔄 Real-time webhook notifications for target price triggers and stop-losses.
- 🔄 Automated limit order placement for Tactical Vault rebalancing.

Phase 3: Advanced Options & Volatility Surface Modeling (Q3 2027)

- 📅 Live Options Chain ingestion (Delta, Gamma, Vega, Theta).
- 📅 Real-time Zero-DTE (0DTE) Gamma exposure alerts.
- 📅 Automated covered-call and protective-put hedging strategies for Core holdings.

Phase 4: Mobile Native Apps & Institutional Multi-Tenancy (Q4 2027)

- 🗓️ React Native iOS and Android applications with biometrics.
- 🗓️ Multi-user roles for RIA advisory firms and family office reporting.
- 🗓️ Automated tax-loss harvesting engine for US & Indian capital gains.

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10. Regulatory, Compliance & Risk Management

1. **Investment Advice Disclaimer:**

- The platform is strictly an **algorithmic research and information tool**, not a registered investment advisor (RIA) or broker-dealer.
- Prominent disclaimers must appear on all trade modals and algo recommendation cards stating: "*Past algorithmic performance is no guarantee of future returns. Invest at your own risk.*"

1. **SEBI & SEC Compliance:**

- Adhere to SEBI algorithmic trading guidelines in India and SEC/FINRA non-advisory technology vendor provisions in the United States.

1. **Data Provider Attribution:**

- Appropriate attribution and compliance with Yahoo Finance, SEC EDGAR, and macroeconomic public API terms of service.